

Exercise Sheet 2

System Safety

Introduction to Machine Learning Safety

System Context

This report applies System-Theoretic Process Analysis (STPA) to the CARLA autonomous driving prototype described in the course roadmap. The system consists of a single forward-facing RGB camera, three binary perception classifiers, a distance oracle, a rule-based planning module, drive-by-wire actuators, and a human safety operator. The three perception models output whether a pedestrian, vehicle, or traffic light is present. The planner uses those outputs, together with speed and steering information, to decide whether to continue driving, decelerate, or emergency brake.

The safety analysis is intentionally conservative. The design has no radar, LiDAR, ultrasonic sensing, or additional safety mechanism. The human operator is the only fallback if automation fails, and the operator may react late or miss an intervention during prolonged monitoring.

Exercise 2.1: Safety Vocabulary

Term	Definition	Autonomous-driving example
Loss	An unacceptable adverse consequence involving harm, damage, legal violation, or environmental impact.	A pedestrian is injured or killed by the autonomous vehicle.
Hazard	A system state or condition that can lead to a loss under worst-case environmental conditions.	The vehicle continues moving while a pedestrian is in its path within stopping distance.
Risk	The combination of likelihood and severity of a hazardous outcome.	A missed pedestrian detection at 50 km/h has high severity and must be treated as high risk even if it is not frequent.

STPA Step 1: Purpose, ODD, Losses, and Hazards

Exercise 2.2: ODD Specification

The Operational Design Domain (ODD) defines the conditions under which the safety claims for this prototype are meaningful. Since the perception models were trained exclusively on sunny daytime CARLA data, the ODD must be narrow. The system should not silently continue autonomous operation outside that boundary.

Dimension	Operating conditions	Non-operating conditions	Runtime ODD-violation detection
Weather	Dry road, clear or mildly cloudy weather, no precipitation, no dense atmospheric obstruction.	Heavy rain, fog, snow, wet windshield, spray, low visibility.	Camera visibility checks, rain/fog classifier, windshield/lens quality monitor, OOD detector.
Lighting	Daytime with sufficient natural illumination and acceptable contrast.	Night, dusk/dawn with low visibility, strong glare, low sun angle, tunnel-like darkness.	Brightness histogram, glare detector, contrast check, OOD detector.
Camera condition	Single forward-facing RGB camera, clean lens, fixed rigid mounting behind windshield, stable 10 Hz image stream.	Dirty lens, occluded lens, defocus, motion blur, displaced camera, dropped frames.	Image sharpness and occlusion checks, calibration consistency check, frame-rate monitor.
Scene type	Public urban roads and mapped intersections similar to CARLA training scenes.	Highways, rural roads, construction zones, unmapped or unusual intersections, off-road scenes.	Map matching, GPS/localization consistency, scene classifier, route/geofence check.
Vehicle speed	0–50 km/h, including slow movement and stops at intersections.	Speeds above 50 km/h or speeds for which stopping-distance assumptions are invalid.	Vehicle speed sensor and planner speed-threshold check.

ODD rule: autonomous mode must either remain inside these conditions or transition to a safe fallback. Continuing autonomous operation outside the ODD is itself hazardous.

Exercise 2.3: Losses

ID	Loss description	Why unacceptable
L-1	Injury or death of a pedestrian or other vulnerable road user.	Direct human harm is the most severe unacceptable outcome.
L-2	Collision with another vehicle.	Can cause injuries, fatalities, property damage, traffic disruption, and legal liability.
L-3	Traffic-law violation, especially at intersections or pedestrian crossings.	Creates unsafe behavior, undermines deployment legitimacy, and may lead to fines or crashes.

L-4	Loss of vehicle control or unstable vehicle behavior.	Can escalate into severe collisions or secondary accidents.
L-5	Damage to public or private infrastructure/property.	Causes economic loss and can endanger road users nearby.

Exercise 2.4: Hazards

ID	Hazard	Loss(es)	Likelihood	Severity
H-1	Vehicle moves while a pedestrian is in its path or within stopping distance.	L-1	Medium	High
H-2	Vehicle fails to stop when stopping is required at an intersection or traffic light.	L-1, L-2, L-3	Medium	High
H-3	Emergency braking occurs unnecessarily or unexpectedly.	L-2, L-4, L-5	Medium	Medium
H-4	Autonomous mode remains active outside the ODD.	L-1, L-2, L-4	High	High
H-5	Human operator fails to intervene when automation is unsafe.	L-1, L-2, L-4	Medium	High
H-6	Vehicle continues toward another road user within speed-dependent stopping distance.	L-1, L-2	Medium	High

STPA Step 2: Control Structure

Exercise 2.5: Control Structure Diagram

The control structure contains two controllers: the planning module and the human safety operator. The perception models and distance oracle are treated as feedback providers, not controllers. This distinction matters: the classifiers do not decide the vehicle action; the planner does.

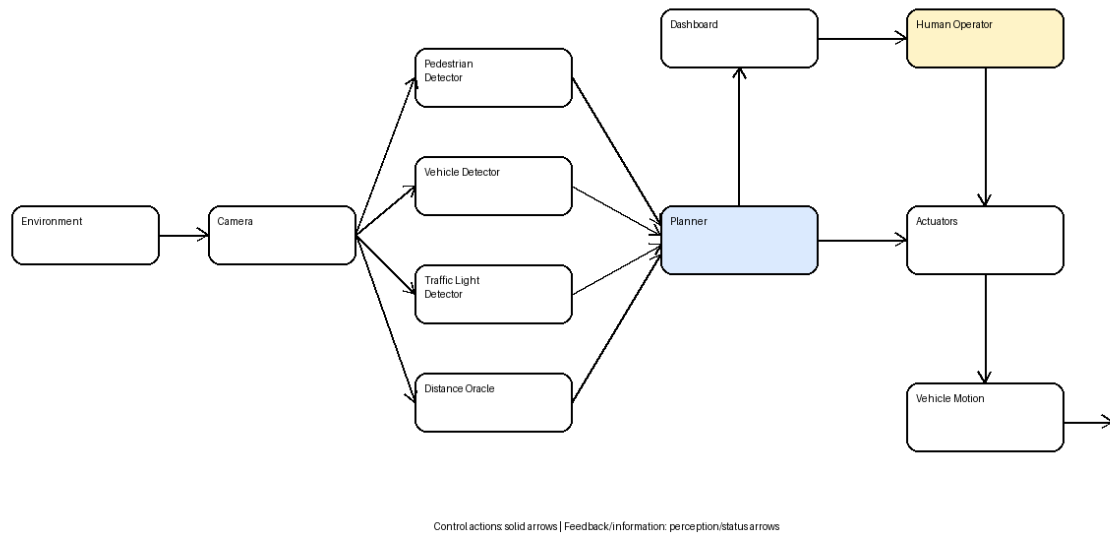


Figure 1: Control structure for the CARLA autonomous driving prototype. Solid arrows represent control actions; dashed arrows represent feedback.

Controllers

- Planning module: commands continue driving, deceleration, or emergency braking.
- Human safety operator: can override with brake or steering input.

Feedback sources

- Camera images at 10 Hz.
- Pedestrian-present, vehicle-present, and traffic-light-present classifier outputs.
- Distance estimates for detected pedestrians and vehicles.
- Vehicle speed and steering angle.
- Dashboard information for the human operator.

Critical limitation: the traffic-light classifier only detects whether a traffic light is present. It does not classify red, yellow, or green. Therefore, the planner cannot safely infer traffic-light state from that model alone. A conservative intersection policy is required.

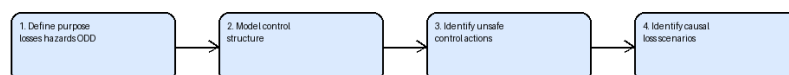


Figure 2: STPA workflow used to move from losses and hazards to unsafe control actions, constraints, and causal scenarios.

STPA Step 3: Unsafe Control Actions and Safety Constraints

Exercise 2.6: Unsafe Control Actions

ID	Controller	Control action	UCA type	Hazard(s)	Unsafe scenario
UCA-1	Planner	Continue driving	Provided unsafely	H-1, H-6	Vehicle continues while a pedestrian or vehicle is within stopping distance.
UCA-2	Planner	Emergency brake	Not provided	H-1, H-6	Planner fails to brake for a detected road user within critical distance.
UCA-3	Planner	Emergency brake	Wrong timing/order	H-1, H-6	Brake command is issued too late to stop safely.
UCA-4	Planner	Continue driving	Wrong duration	H-2	Vehicle continues through an intersection when stopping is required or traffic-light state is unknown.
UCA-5	Planner	Emergency brake	Provided unsafely	H-3	False-positive perception causes unnecessary maximum braking.
UCA-6	Human operator	Manual brake override	Not provided	H-5	Operator does not intervene when automation misses a critical hazard.
UCA-7	Human operator	Steering or brake override	Wrong timing/order	H-4, H-5	Operator reacts too late after an ODD violation or unsafe planner action.
UCA-8	Human operator	Manual override	Wrong duration	H-4, H-5	Operator releases control before the vehicle reaches a safe state.

Exercise 2.7: Safety Constraints

UCA	Safety constraint	Level	Verification
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UCA-1	The planner must not continue driving when a pedestrian or vehicle is within stopping distance.	System-level	Closed-loop scenario testing with road users at different distances and speeds.
UCA-2	The planner must issue emergency braking when a road user is inside the critical distance threshold.	System-level	Simulation tests and rule-based planner tests.
UCA-3	End-to-end perception-to-braking latency must remain below the maximum safe response time.	Model-level and system-level	Inference latency benchmark and closed-loop timing test.
UCA-4	At intersections, the planner must behave conservatively when traffic-light state is unknown or ambiguous.	System-level	Intersection scenario tests and requirement review.
UCA-5	Emergency braking must not rely on a single unreliable positive signal without plausibility checks where unnecessary braking creates risk.	System-level	False-positive and following-vehicle scenario tests.
UCA-6	The operator must receive clear and timely information during critical automation failures.	System-level	Human-factor evaluation and dash-board/warning review.
UCA-7	The system must detect ODD violations and initiate fallback before relying solely on late manual takeover.	System-level	OOD detection and fallback transition tests.
UCA-8	Manual takeover must remain active until the vehicle has reached a defined safe state.	System-level	Operator protocol review and takeover simulation.

STPA Step 4: Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

UCA	Causal scenario	Root cause	Related constraint
UCA-1	Pedestrian detector outputs “no pedestrian” although a pedestrian is in the driving path; planner continues.	False negative caused by lighting, occlusion, unusual pedestrian appearance, or ODD gap.	Planner must not continue near uncertain or critical detections.

UCA-2	Emergency braking is not triggered because the planner receives no positive pedestrian or vehicle signal.	Binary classifier miss; planner fully depends on perception outputs and has no raw-image access.	Planner must brake when road user is inside critical distance; model recall must be sufficient.
UCA-3	Correct detection arrives after the remaining stopping distance is already too short.	Perception or control-loop latency; 10 Hz camera update can be insufficient in fast-changing scenes.	Latency must remain below safe response threshold.
UCA-4	Traffic light is detected as present, but the planner cannot know whether it is red, yellow, or green.	Perception model has wrong abstraction: presence detection is not traffic-state recognition.	Planner must use conservative intersection behavior when state is unknown.
UCA-5	Vehicle detector produces a false positive and planner commands maximum braking.	Classifier false positive, reflection, object confusion, or unusual scene geometry.	Emergency braking requires plausibility checks and risk-aware thresholds.
UCA-6	Operator misses the automation failure and does not brake.	Fatigue, attention lapse, no auditory alert, long monitoring shift.	Operator must receive timely and salient failure information.
UCA-7	Rain or glare begins during the drive but autonomous mode remains active; operator reacts late.	ODD violation not detected early enough; fallback depends too much on human reaction.	System must detect ODD violations and transition to safe fallback.
UCA-8	Operator takes over briefly but releases the brake/steering before the hazardous situation is resolved.	Mode confusion, unclear dashboard state, or poor takeover protocol.	Manual override remains active until safe-state criteria are met.

Exercise 2.9: Mapping Constraints to Future Evidence

Constraint or evidence need	Future exercise sheet / evidence source
Baseline pedestrian, vehicle, and traffic-light classifier behavior; initial ODD gap analysis.	Sheet 3: train three baseline models and compare intended ODD with data.
Precision, recall, F1, and confusion matrices for each perception model.	Sheet 4: Testing I.

ODD coverage, ODD refinement, and distribution-shift analysis.	Sheet 5: Testing II.
Robustness of perception models under adversarial perturbations.	Sheet 6: Adversarial ML.
Calibration, confidence reliability, and uncertainty evidence.	Sheet 7: Uncertainty.
Evidence that classifiers focus on relevant image regions.	Sheet 8: Explainability with GradCAM.
Ability to detect rain, fog, night, or other out-of-distribution inputs.	Sheet 9: Anomaly/OOD detection.
Final argument connecting constraints, evidence, assumptions, and residual risk.	Sheet 10: Complete safety case.

Safety Assessment Summary

The current design is not robust enough for dependable public-road autonomy. The main weaknesses are structural, not cosmetic: one camera, binary perception outputs, no sensor redundancy, no independent safety monitor, no auditory warning, and a human operator as the only fallback. The traffic-light detector is especially weak for planning because detecting traffic-light presence is not the same as detecting whether stopping is required.

The minimum defensible safety argument must therefore rely on strict ODD enforcement, conservative fallback behavior, high recall for pedestrian and vehicle detection, bounded latency, OOD detection, clear operator takeover rules, and explicit evidence from later exercise sheets.